Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/25/2005
PI Name:	Cohen, Richard J. M.D., Ph.D.		
Project Title:	Effects of Simulated Microgravity on Cardiovascular Stability		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rjcohen@mit.edu	**Fax:**	FY 617-253-3019
PI Organization Type:	UNIVERSITY	**Phone:**	617-253-7430
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	45 Carleton Street Rm. E25-335A		
PI Address 2:			
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02142-1323	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	End date changed from3/31/2008.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Grenon, Marlene (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-CA00402		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The purpose of this investigation is to evaluate and test countermeasures to the development of microgravity induced orthostatic intolerance and susceptibility to ventricular dysrhythmias. We will apply the Cardiovascular System Identification (CSI) technique for assessing closed-loop cardiovascular regulation and Microvolt T-Wave Alternans (MTWA) technique for assessing susceptibility to ventricular dysrhythmias. The specific aims of the project are: 1. To test a panel of countermeasures to orthostatic intolerance in subjects intolerant to pre bed rest head-up tilt, in conjunction with the evaluation of CSI and other cardiovascular measures as a potential means of predicting orthostatic intolerance and countermeasure effectiveness. 2. To test in a 16 day head-down tilt bed rest study the effectiveness of the orthostatic intolerance countermeasure identified for each individual during pre bed rest screening, in conjunction with the evaluation of CSI and other cardiovascular measures as a means of predicting orthostatic intolerance and		

	countermeasure effectiveness. 3. To utilize MTWA analysis, in older men and post-menopausal women, to measure the effects of a panel of countermeasures for reducing cardiac electrical instability, and to study the effects of these countermeasures on baseline orthostatic tolerance and closed-loop cardiovascular regulation as measured by CSI and other cardiovascular measures. 4. To test utilizing MTWA analysis in a 16 day head-down tilt bed rest study the effectiveness of the cardiac electrical instability countermeasure identified for each individual during pre bed rest screening. 5. To further develop and enhance the methodology of CSI, MTWA and Plethysmography analyses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has multiple earth based applications: Specifically, the research has direct application to the study of mechanisms of, and treatment for, orthostatic hypotension her on earth. Orthostatic hypotension is an an important and prevalent clinical problem which can lead to syncope and falls, which particularly in the elderly, may be life threatening. In addition, this study also focuses on mechanisms and treatment of life threatening ventricular dysrhythmias. Sudden cardiac death from ventricular dysrhythmias accounts for one-half of all cardiac deaths and one in seven of all deaths. The technologies developed in the context of this project all have direct application to earth based medicine. The microvolt T-wave alternans testing technique has already been successfully commercialized, cleared by the FDA, is reimbursed for by Medicare, and is in widespread clinical used to identify patients at high and low risk of sudden cardiac death sot that they may be appropriately treated. Cardiovascular system identification is a technology that has potential widespread applications for a variety of disease states such as heart failure, hypertension, diabetes, etc which involve alterations in cardiovascular regulation. Continuous cardiac output monitoring is a technology that may have widespread clinical importance particularly in intensive care type situations. It may be particularly important for combat casualty care. Mathematical analysis of venous plethysmography data may play an important role in diagnosing mechanisms of orthostatic intolerance.
Task Progress:	This past year we concluded the enrollment of subjects in our study of the effects of 16 day head-down tilt bed rest on orthostatic intolerance in premenopausal women and the effectiveness of the midodrine countermeasure. The data from this study are now being analyzed. We initiated enrollment of subjects in our protocol to evaluate a panel of countermeasures to orthostatic intolerance after 16 day head down tilt bed rest. We have analyzed much of the data from our bed rest studies to date, as well as analyzed other animal and clinical data in order to validate the new technologies which we have developed for the purposes of these studies. These technologies include microvolt T-wave alternans testing to assess non-invasively susceptibility to life threatening ventricular dysrhythmias, cardiovascular system identification to assess non-invasively autonomic function and closed-loop hemodynamic regulation, continuous cardiac output monitoring from analysis of the arterial blood pressure waveform, and novel quantitative analysis of calf plethysmographic data. The detailed findings from these analyses are described in the eight peer reviewed journal article publications emanating from this past year's work and in the section of this report entitled Main Findings.
Bibliography Type:	Description: (Last Updated: 01/18/2008)
Articles in Peer-reviewed Journals	Aljuri N, Marini R, Cohen RJ. "Test of dynamic closed-loop baroreflex and autoregulatory control of total peripheral resistance in intact and conscious sheep." Am J Physiol Heart Circ Physiol. 2004 Nov;287(5):H2274-86. Epub 2004 Jul 1. PMID: 15231504 , Nov-2004
Articles in Peer-reviewed Journals	Armoundas AA, Albert CM, Cohen RJ, Mela T. "Utility of implantable cardioverter defibrillator electrograms to estimate repolarization alternans preceding a tachyarrhythmic event." J Cardiovasc Electrophysiol. 2004 May;15(5):594-7. PMID: 15149432 , May-2004
Articles in Peer-reviewed Journals	Grenon SM, Hurwitz S, Sheynberg N, Xiao X, Judson B, Ramsdell CD, Kim C, Cohen RJ, Williams GH. "Sleep restriction does not affect orthostatic tolerance in the simulated microgravity environment." J Appl Physiol. 2004 Nov;97(5):1660-6. Epub 2004 Jul 2. PMID: 15234956 , Nov-2004
Articles in Peer-reviewed Journals	Grenon SM, Sheynberg N, Hurwitz S, Xiao G, Ramsdell CD, Ehrman MD, Mai CL, Kristjansson SR, Sundby GH, Cohen RJ, Williams GH. "Renal, endocrine, and cardiovascular responses to bed rest in male subjects on a constant diet." J Investig Med. 2004 Mar;52(2):117-28. PMID: 15068228 , Mar-2004
Articles in Peer-reviewed Journals	Grenon SM, Hurwitz S, Sheynberg N, Xiao X, Ramsdell CD, Mai CL, Kim C, Cohen RJ, Williams GH. "Role of individual predisposition in orthostatic intolerance before and after simulated microgravity." J Appl Physiol. 2004 May;96(5):1714-22. PMID: 15075309 , May-2004
Articles in Peer-reviewed Journals	Hurwitz S, Cohen RJ, Williams GH. "Diurnal variation of aldosterone and plasma renin activity: timing relation to melatonin and cortisol and consistency after prolonged bed rest. " J Appl Physiol. 2004 Apr;96(4):1406-14. Epub 2003 Dec 5. PMID: 14660513 , Apr-2004
Articles in Peer-reviewed Journals	Xiao X, Mukkamala R, Sheynberg N, Grenon SM, Ehrman MD, Mullen TJ, Ramsdell CD, Williams GH, Cohen RJ. "Effects of simulated microgravity on closed-loop cardiovascular regulation and orthostatic intolerance: analysis by means of system identification. " J Appl Physiol. 2004 Feb;96(2):489-97. Epub 2003 Sep 26. PMID: 14514703 , Feb-2004
Articles in Peer-reviewed Journals	Aljuri N, Cohen RJ. "Theoretical considerations in the dynamic closed-loop baroreflex and autoregulatory control of total peripheral resistance." Am J Physiol Heart Circ Physiol. 2004 Nov;287(5):H2252-73. Epub 2004 Jul 1. PMID: 15231503 , Nov-2004
Awards	Grenon M. "Marlene Grenon was named 1st Runner-Up in the Space Medicine Branch Young Investigator Award competition for her presentation at the 2004 Aerospace Medical Association in Anchorage, AK. 5/10/04." May-2004
Awards	Grenon M. "Marlene Grenon won the AFMR Henry Christian Award at the Clinical Research meeting in Chicago, IL. 4/16/04." Apr-2004
Patents	Pending 2004. Pending Patent, 2004. Apr-2004 Mukkamala R, Sherman D, Cohen, RJ. "METHODS AND APPARATUS FOR DETERMINING CARDIAC OUTPUT."

Presentation	Bheemreddy SR, Grenon SM, Xiao X, Hurwitz S, Sheynberg N, Kim C, Cohen RJ, Williams GH "Cardio-Endocrine, Renal Responses and Influence of Gender in Humans to Simulated Microgravity" N/A Jan-2005
Presentation	Grenon SM, Xiao X, Sheynberg N, Hurwitz S, Cohen RJ, Williams GH "The Effect of Sleep Restriction on Orthostatic Intolerance in the Simulated Microgravity Environment" N/A Oct-2004
Presentation	Grenon SM, Xiao X, Sheynberg N, Hurwitz S, Cohen RJ, Williams GH "The Role of Individual Predisposition in Orthostatic Intolerance Before and After Simulated Microgravity" N/A May-2004
Presentation	Grenon SM, Xiao X, Sheynberg N, Hurwitz S, Kim C, Williams GH, Cohen RJ "Simulated Microgravity Induces Microvolt T-Wave Alternans" N/A Oct-2004
Presentation	Grenon SM, Xiao X, Sheynberg N, Hurwitz S, Kim C, Williams GH, Cohen RJ. "Simulated Microgravity Induces Microvolt T-Wave Alternans" N/A Oct-2004
Presentation	SM Grenon, RJ Cohen, C Kim, S Hurwitz, CD Ramsdell, X Xiao, N Sheynberg, and GH Williams "A New Model for the Role of Individual Predisposition in Orthostatic Intolerance Before and After Exposure to Simulated Microgravity" N/A Apr-2004
Presentation	Xiao X, Grenon SM, Kim C, Williams GH, Cohen RJ "Cardiovascular Effects of Simulated Microgravity" N/A Jan-2005